

Yashashwani Garg

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EDUCATION

Indian Institute of Science Education and Research Berhampur

2021 - 2026

Currently in fifth year

CPI: 9.3

FIITJEE Junior College

2019 - 2021

Higher Secondary Education

Percentage score: 98.2

ACADEMIC PROJECTS and WORKSHOPS

Masters' Project

June 2025 - Present

IISc Bangalore

Mentor: Prof Aweek Bid

- We worked on exfoliation of $MoTe_2$ monolayers and their characterization using Raman Spectroscopy and Photoluminescence analysis
- We are working on fabrication of Moiré graphene devices and measurement of their thermodynamic compressibility using quantum capacitance measurements

Summer Internship

May 2024 - July 2024

IIT Kanpur

Mentor: Prof Amit Kumar Agarwal

- I performed literature survey on the basics of superconductivity, the Josephson Effect and the Superconducting Diode Effect.
- As part of the internship, I did elementary calculations of transmission coefficients for Symmetric and asymmetric rectangular potential barriers and multiple potential barriers and also replicated the Transmission coefficient calculation for single and bi-layer Graphene.

Summer Internship

May 2023 - July 2023

TIFR Mumbai

Mentor: Prof Arumugam Thamizhavel

- I grew single crystals of $CeIn_3$ and $LaIn_3$ using the flux growth method
- The magnetic and electronic properties of the crystals grown were studied and the trends in their magnetic susceptibility and resistivity as a function of applied magnetic field were replicated.

Reading Project

December 2022

IISER Berhampur

Mentor: Dr. Rahul Sharma

- I studied the basics of Solid State Physics and introductory courses to Superconductivity.

Intro2Astro

May 2022 – June 2022

Online Workshop

Mentors/ Organizers: Dr. Fei Dai, Dr. Isaac Howardson

- I gained knowledge and experience in using Python programming language, covering plotting of data and its analysis.
- I was introduced to various soft skills including scientific paper reading, web development and CV creation.

Pyar (Python and Research) Tutorial

January 2022

University of California, Santa Cruz

Mentors/ Organizers: UCSC

- An online workshop that introduced me to the basics of python programming using the Jupyter notebook.

EXTRA-CURRICULAR ACTIVITIES

- I worked as a member of the Events team and contributed towards a smooth speaker-institute communication for TEDxIISERBPR'24, a TEDx event organized by our institute.
- I worked as the lead of the PR Team of IISER Berhampur's Entrepreneurship Cell (IIEC), from February 2022 to March 2023.
- During my tenure I organized three talks focussing on entrepreneurship and the opportunities available to students after the completion of their BS-MS studies.
- I placed second in a case-challenge event, Enigma, conducted by the E-Cell of IIT Bhubaneswar during E-Summit, IIT Bhubaneswar in February, 2023

ADDITIONAL INFORMATION

I have working knowledge in

- Python
- MATLAB
- LaTeX Overleaf
- ORIGIN data analysis and graphing